

Ai4Energy

Anomaly and Outlier Detection for Temperature Measurement Channels

Approach guidance for the software development team

Scope: Detecting faulty, corrupted, and physically impossible temperature readings when no historical baseline and no labelled anomaly data exist - the situation for every new experiment.

Hardware in scope: K-type thermocouples via MAX31855 and MAX31856; DS18B20 (1-Wire); SHT45; DHT11. Keysight DAQ970A with 34901A multiplexer as the validation reference instrument.

Audience: Software engineers implementing the detection logic. No prior heat-transfer background assumed.

What this document is: A description of how to think about the problem and which tests to build, with the physical reasoning behind each. It deliberately contains no source code - the algorithms are stated as principles and equations so the team can implement them in whichever language and architecture suits the node and cloud layers.

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Contents

If the list below appears empty, click anywhere in it and press F9 (or right-click and choose Update Field) to populate it.

Contents	2
1. The problem, stated precisely	3
1.1 Why the usual approach does not work here	3
1.2 The reframe: the physics is the reference model	3
1.3 The three-way classification	3
1.4 The cardinal rule: annotate, never filter	4
2. The measurement chain and where it breaks	5
2.1 Front-end comparison	5
2.2 Layer 0 - signal integrity checks	5
2.3 Timing and scheduling constraints that affect detection	6
3. The physics as the reference model	8
3.1 The sensor as a dynamical system	8
3.2 Why τ is the most valuable prior available	9
3.3 The measurement is not the gas temperature	10
3.4 The K-type thermocouple as a transducer	11
4. Turning the physics into detection tests	14
4.1 Amplitude bounds from thermodynamics	14
4.2 Rate bounds and the sensor as a low-pass filter	14
4.3 The two-sensor lag check	15
4.4 Measuring τ in situ, and reading its drift	15
4.5 Trajectory and energy-balance sanity	15
4.6 A statistical layer, correctly bounded	16
5. Validating against the DAQ970A dataset	18
5.1 NPLC integration makes the instrument a different low-pass filter	18
5.2 Accuracy versus repeatability	18
5.3 Practical checklist for using the validation set	19
6. Summary: physics to detector	20
Suggested implementation order	20
Items to verify rather than assume	21

1. The problem, stated precisely

1.1 Why the usual approach does not work here

The standard formulation of anomaly detection is: collect a large body of data representing normal operation, learn a model of it, then flag whatever deviates. That formulation is unavailable to this project, for three independent reasons.

- **No history.** A new experiment has, by definition, never been run before. There is no corpus of previous runs for that configuration to learn from. The detector must work correctly from the first minute of the first run.
- **No labels.** Nobody has marked which past samples were faulty. Without labels there is no way to measure a false-negative rate, and therefore no way to validate a learned detector - you cannot know whether it works.
- **No stationary "normal".** A research rig's operating distribution is changed deliberately, every run. That is what an experiment is. Any model that learns "normal" from past runs will flag every novel condition - and the novel conditions are the entire point of the facility.

A fourth, subtler failure mode matters just as much. A model trained on a rig that has been quietly miscalibrated since commissioning will learn the miscalibration as normal, and will then never report it. Learned baselines absorb pre-existing faults.

1.2 The reframe: the physics is the reference model

What the project lacks in historical data, it more than makes up for in prior knowledge. Two things are known before a single sample is taken:

1. **What the sensor is.** The datasheet specifies its range, resolution, accuracy envelope, conversion timing, fault-reporting behaviour, and failure signatures. These are hard facts, available at compile time.
2. **What physics the measurement must obey.** A thermocouple bead is a physical object with mass and heat capacity, coupled to its surroundings by convection, radiation and conduction. Those couplings impose mathematical constraints on what a genuine signal can do: how fast it can change, what frequency content it can contain, how two sensors watching the same process must relate, and what a cooling curve must look like.

The organising principle. A signal that violates the physics of heat transfer to the sensor is provably a measurement fault - it is not an unusual temperature, it is not a temperature at all. A signal that obeys the physics but surprises you may be the science. The detector's job is to separate these two, and never to suppress the second.

This is a stronger position than a learned baseline, not a weaker one. Physical constraints are exact, need no training data, are auditable, and can be stated in a publication: "data were flagged under quality ruleset v1.3, with thresholds derived from IEC 60584 Class 2 tolerances and the measured thermal time constant of each probe." That sentence cannot be written about a continuously adapting neural model.

1.3 The three-way classification

Every flagged sample should fall into one of three classes, and the classes carry different authority. Building this distinction into the architecture from the start prevents the characteristic failure of anomaly detection in research settings, which is laundering discoveries out of the dataset.

Class	Definition	Typical examples	Permitted action
Implausible (physics-violating)	Breaks a hard physical or datasheet constraint even after generous tolerance	Rate of change faster than the sensor's thermal time constant permits; reading above the adiabatic flame temperature; hardware fault bit set; failed CRC	May be excluded from the science dataset. Raw value still retained.
Suspicious (instrument signature)	Physically consistent, but matches a known degradation pattern	Fitted time constant drifting upward; bias growing with soot deposition; step change after a maintenance window	Annotate and raise as a maintenance diagnostic. Data retained.
Unusual (novel but physical)	Obeys every physical constraint but is statistically surprising	An unexpected excursion that respects the sensor's dynamics; oscillation within the thermal passband, which may be genuine combustion instability	Surface to the researcher. Never suppress - this may be the result.

Architectural rule. Physics-based checks may veto data. Statistical checks may only annotate it. Keeping that asymmetry explicit in the code prevents statistical machinery from silently deleting real phenomena.

1.4 The cardinal rule: annotate, never filter

The node and the pipeline must always transmit the raw measured value, accompanied by a quality-flag field. They must never substitute, clamp, smooth, or drop a reading on the basis of a detection result.

The reason is asymmetric risk. If a bad sample is passed through with a flag, a researcher can filter it later, at leisure, with full context. If a good sample is deleted at the node, it is gone permanently - and the samples most likely to trip a naive detector are precisely the fastest, most extreme transients, which in a combustion experiment are usually the most scientifically valuable five seconds of the run.

2. The measurement chain and where it breaks

Before any physics can be applied, the software must establish that the number it received is a number the hardware actually produced. This is a distinct layer from anomaly detection proper, and it must come first - every physical test below assumes the bytes are real.

2.1 Front-end comparison

The three K-type acquisition paths in this project have materially different properties, and the detection logic must be parameterised per front-end, not per sensor type. Two identical probes read by a MAX31855 and a MAX31856 will legitimately produce different noise character and slightly different levels.

Property	MAX31855	MAX31856	DAQ970A + 34901A
Linearisation	None - assumes a constant 41.276 $\mu\text{V}/^{\circ}\text{C}$. Error grows toward range extremes.	Internal lookup table, ITS-90 corrected, for K/J/N/R/S/T/E/B.	Full instrument-grade linearisation.
Integrity check on the data word	None. No CRC, no parity.	Register-based; fault status register with 8 defined fault bits.	Instrument-grade; SCPI error queue.
Fault detection	Open circuit, short to VCC, short to GND.	Open circuit, over/under-voltage, thermocouple out-of-range / high / low, cold-junction out-of-range / high / low.	Open-transducer detection; configurable limits.
Mains rejection	None.	Built-in 50/60 Hz filter.	NPLC integration - rejects mains by construction.
Input protection	Standard.	$\pm 45\text{ V}$ on thermocouple inputs.	Instrument-grade, relay multiplexed.
Resolution (LSB)	0.25 $^{\circ}\text{C}$	$\approx 0.0078\text{ }^{\circ}\text{C}$ (19-bit register)	6½ digits (22-bit) in voltage mode
Role in this project	Existing low-cost nodes	Preferred for new nodes	Validation reference only

The single most important line in that table. The MAX31855 SPI word carries no CRC and no parity. A floating MISO line, a stuck chip-select, or a bus conflict returns all-zeros - which decodes as a perfectly plausible 0.00 $^{\circ}\text{C}$ thermocouple reading, 0.00 $^{\circ}\text{C}$ cold junction, and no fault bit set. This is a silent, physically reasonable, completely wrong value, and no amount of statistical anomaly detection will catch it in a rig operating near ambient. It must be caught structurally.

2.2 Layer 0 - signal integrity checks

These cost almost nothing and catch failure modes that no downstream statistical method can recover from.

MAX31855

- Reject an all-zeros or all-ones word outright. Neither is a legitimate reading in any real installation.
- The word contains reserved bits that always read as zero. If a reserved bit is set, the word is corrupt - this is a free integrity check standing in for the absent CRC. [Verify the exact reserved bit positions against the current datasheet before implementing.]
- The overall fault bit must be consistent with the three specific cause bits. If the summary bit and the cause bits disagree, the transfer is corrupt rather than the sensor being faulty.

- Optionally read twice a few milliseconds apart. The part free-runs at roughly 100 ms per conversion, so two closely spaced reads should be bit-identical; a difference indicates bus trouble.

MAX31856

- Read and decode the fault status register every acquisition cycle. It reports eight distinct conditions - open thermocouple, over/under-voltage, and out-of-range, high and low conditions for both the thermocouple and the cold junction. Map each to a distinct quality flag rather than collapsing them into a single "fault" bit; the specific cause is diagnostically valuable.
- Because this part is register-addressed rather than a single free-running word, verify register readback consistency on configuration registers periodically. A device that has silently reset will have reverted to default configuration - including a different thermocouple type - and will produce plausible but wrong readings.
- The over/under-voltage fault is particularly valuable in a combustion rig: it detects the input being driven outside the normal thermocouple range, which is the signature of igniter EMI coupling or a wiring fault, and it is not inferable from the temperature data alone.

DS18B20 and the 1-Wire bus

- **Always validate the scratchpad CRC-8.** This is the primary integrity mechanism and it is mandatory, not optional.
- **Do not rely on sentinel values.** The commonly cited -127°C is not a device output at all; it is a library constant from the Arduino DallasTemperature package. The genuine bus-failure signatures are a raw all-ones word (no device responding, line pulled up) and a raw all-zeros word (line held low). The latter decodes as 0.0°C , which is entirely plausible in a rig operating near ambient - another silent failure that only the CRC and the presence pulse will catch.
- **Address devices by their unique 64-bit ROM ID, never by discovery order.** If channels are indexed by enumeration order and one device drops off the bus mid-run, every downstream channel label silently shifts - the reactor inlet becomes the reactor outlet, with nothing flagged. Store an expected ROM-ID manifest, verify it at start-up and periodically during the run, and treat any change in bus topology as a first-class anomaly.
- **Handle the 85.0°C power-on value carefully.** The temperature register powers up at exactly $+85.0^{\circ}\text{C}$ and holds that value until a conversion completes. But 85.0°C is also a legitimate temperature. Flag it only when the exact power-on code is present and no conversion completed in that cycle - do not blanket-reject the value.
- **Track the CRC failure rate per device as a telemetry channel.** A rising CRC error rate is the earliest available warning of a degrading connection or a marginal bus, and it costs one counter per device.
- **Prefer external three-wire power over parasite power.** Parasite operation imposes strict timing on the strong pull-up and cannot supply many devices converting simultaneously; it removes an entire class of intermittent failures for the cost of one conductor.
- **Be aware that counterfeit DS18B20 parts are widespread.** Some ignore resolution configuration, some share duplicate ROM IDs - which corrupts bus enumeration in ways that present as random dropouts. A one-time commissioning check against published identification methods is worthwhile.

2.3 Timing and scheduling constraints that affect detection

Several practical issues will corrupt the physics-based tests downstream if they are not handled at acquisition time.

- **Timestamp from a monotonic hardware source at the instant of the read.** Every test that involves a rate of change divides by the sample interval. If the code assumes one second and network activity delayed the read to 1.35 s, the computed slope is wrong by 35 % and the rate test false-alarms. Wall-clock time must never be used for this: a clock synchronisation step can produce a zero, negative, or enormous interval and will destroy any filter that divides by it. Use wall clock only for the transmitted timestamp.
- **Account for the DS18B20 conversion time.** A 12-bit conversion takes about 750 ms, which cannot be blocked on within a one-second cycle alongside network activity. The standard solution is to command the conversion on one cycle and read the result on the next - which makes DS18B20 data one sample older than the thermocouple data. Any cross-sensor comparison must account for that offset explicitly, or it will manufacture phantom lag anomalies. Reducing to 11-bit resolution halves the conversion time and is usually the better trade.
- **Persist detector state across restarts.** Drift accumulators and fitted parameters must survive a reboot. A node that browns out on a transmission current spike and resets will otherwise silently zero its drift monitor. Stamp every record with a boot identifier and monotonic uptime so the pipeline can detect resets and data gaps.
- **Consider oversampling the thermocouple front-end.** If the converter produces readings faster than the transmission rate, compute and publish the mean, standard deviation, minimum and maximum over each transmitted interval rather than a single instantaneous value. The within-interval standard deviation is the best single diagnostic available for electrical interference and loose terminal connections - both show as elevated short-timescale scatter long before they distort the mean.

3. The physics as the reference model

This section supplies the physical reasoning that the detection tests in Section 4 are built on. It is written for implementers rather than heat-transfer specialists, but the equations matter - they are what make the tests quantitative rather than arbitrary.

3.1 The sensor as a dynamical system

Treat the sensing element - the thermocouple bead, or the DS18B20 die and package - as a single lump of material at uniform temperature T , immersed in a fluid at temperature T_∞ , exchanging heat by convection. The energy balance is:

$$\rho V c \cdot dT/dt = h A (T_\infty - T)$$

where ρ is the material density, V its volume, c its specific heat capacity, A the surface area exchanging heat, and h the convective heat transfer coefficient - which is a property of the flow, not of the sensor. Rearranged:

$$dT/dt = (T_\infty - T) / \tau \quad \text{with} \quad \tau = \rho V c / (h A)$$

This is a first-order linear system. For a step change in the surrounding temperature the solution is the familiar exponential approach:

$$T(t) = T_\infty + (T_0 - T_\infty) \cdot e^{(-t/\tau)}$$

τ is the thermal time constant - the time taken to close 63.2 % of the remaining gap to the final temperature. After 3τ the sensor is within 5 % of final value, after 5τ within 1 %. This single parameter is the most useful physical prior available to the detector.

What physically determines τ

For a spherical bead of diameter D , the volume-to-area ratio is $D/6$, giving $\tau = \rho c D / (6h)$. Three factors move it:

- **Size.** τ scales with bead diameter, and in practice faster than linearly, because h also falls as D grows. This is why fine-wire thermocouples exist.
- **Thermal mass per unit area.** Chromel and alumel have a volumetric heat capacity of roughly $3.5\text{--}4 \times 10^6 \text{ J}/(\text{m}^3 \cdot \text{K})$. A sheath adds stainless steel plus compacted magnesium oxide insulation - far more stored energy per unit of exposed area.
- **The heat transfer coefficient h , which depends entirely on the fluid and the flow.** Still air gives roughly $5\text{--}15 \text{ W}/(\text{m}^2 \cdot \text{K})$ around a large object; flowing combustion gas $50\text{--}500$; stirred liquids $500\text{--}10\,000$. Note an important caveat for small beads: the conduction-limited Nusselt number for a sphere is 2, so $h = 2k_{\text{gas}}/D$ regardless of flow. For a 0.25 mm bead in room air that alone gives $h \approx 200 \text{ W}/(\text{m}^2 \cdot \text{K})$ - small sensors are fast even in still air.

Consequence for configuration. The same probe has a different τ in every installation. τ is a property of the probe and the flow together, not of the probe alone. Moving a probe from a fast gas stream to a recirculation zone, or from gas to liquid, changes τ by one to two orders of magnitude. It must therefore be a per-channel configuration value, ideally measured in situ rather than taken from a catalogue.

When the single-lump model is valid: the Biot number

The lumped model assumes the sensor is internally isothermal - that heat conducts through the sensor much faster than it arrives at the surface. The criterion is the Biot number, $Bi = h \cdot L_c / k_s$, where L_c is the volume-to-area ratio and k_s the conductivity of the sensor body. Below about 0.1 the lumped model

is excellent and the response is a clean single exponential; above it, internal gradients matter and the response becomes a sum of exponential modes.

For a bare metal bead in gas, B_i is very small and the single-exponential model is accurate. For a sheathed probe the assumption fails in a specific way: the heat path runs gas $\rightarrow$ sheath $\rightarrow$ magnesium oxide powder $\rightarrow$ junction, and that powder is a poor conductor. The sheath and the junction behave as two coupled lumps, and the observed step response is better described by two exponentials:

$$T(t) = T_{\infty} + a_1 \cdot e^{(-t/\tau_1)} + a_2 \cdot e^{(-t/\tau_2)} \quad (\tau_1 < \tau_2)$$

Practically, a sheathed probe still behaves as a low-pass system, but with a fast partial response followed by a slow settle. Fitting a single exponential yields a usable effective τ with systematic residuals in the early part of the transient - that is expected physics, and the detector must not flag the fast early portion of a sheathed-probe step as a violation.

Realistic magnitudes

These are order-of-magnitude engineering ranges, to be used as priors for sanity bounds and per-channel defaults. The value for a specific probe in a specific rig must be measured.

Configuration	Typical τ	What moves it within the range
Bare/exposed bead, ~ 0.25 mm, fast hot gas (10–50 m/s)	≈ 10 –100 ms	Gas velocity and temperature, weld bead size
Bare bead, ~ 0.25 mm, still room air	≈ 0.5 –2 s	Draughts, orientation, radiation to surroundings
Sheathed 1.5 mm, ungrounded, flowing gas	≈ 5 –30 s	Gas velocity, junction style, sheath fill
Sheathed 3 mm, ungrounded, flowing gas	≈ 15 –90 s	Grows roughly with sheath diameter
Sheathed 1.5–3 mm, stirred or flowing liquid	≈ 0.3 –5 s	Liquid velocity; grounded junctions at the fast end
DS18B20, TO-92 package, still air	≈ 10 –30 s (package limited)	Airflow and mounting, plus the 750 ms conversion cadence

Two consequences are worth internalising. First, the spread across these configurations is four orders of magnitude, so any rate-of-change threshold must be per-channel and derived from that channel's τ - a single global slope limit is either useless on the fast channel or blind on the slow one. Second, a grounded-junction sheathed probe is typically two to three times faster than the same-diameter ungrounded probe, so changing probe type changes the physical prior.

3.2 Why τ is the most valuable prior available

The first-order model bounds the derivative of any genuine thermal signal:

$$|dT/dt| \leq |T_{\text{drive,max}} - T| / \tau$$

where $T_{\text{drive,max}}$ is the most extreme temperature the environment can present. The sensor cannot change faster than this, whatever the process does - even an instantaneous, infinitely hot flame front produces at most this initial slope, because the bead's own heat capacity must be charged through a finite thermal conductance.

The strongest zero-baseline rule you have. Any recorded change faster than the τ bound is provably non-thermal. In rough order of likelihood for this hardware: SPI or 1-Wire bit corruption (recall the MAX31855 has no CRC, so a flipped bit in the temperature field is silent and can be large), electromagnetic interference coupled into the microvolt-level thermocouple loop, a cold-junction compensation glitch, connector intermittency, an ADC or reference glitch, or a software defect in byte order, sign extension, or buffer handling.

3.3 The measurement is not the gas temperature

A thermocouple reports the temperature of its own junction. In a benign environment that equals the fluid temperature within tolerance. In a combustion rig it does not, and the discrepancies are large, systematic, and - critically - they drift in ways a naive detector will misclassify as faults.

At steady state the heat flows must balance:

$$h A (T_g - T_{tc}) = \epsilon \sigma A (T_{tc}^4 - T_w^4) + Q_{\text{conduction}}$$

Here T_g is the true gas temperature, T_{tc} the junction temperature actually recorded, T_w the effective temperature of the surfaces the bead radiates to, ϵ the emissivity of the bead surface, and σ the Stefan-Boltzmann constant. The junction settles where these balance - generally below the gas temperature in a hot-gas, cold-wall rig, because it radiates heat away and conducts heat out along its own stem.

Radiation error

Neglecting conduction and solving for the error:

$$\Delta T_{\text{radiation}} = T_g - T_{tc} = \epsilon \sigma (T_{tc}^4 - T_w^4) / h$$

As a worked example, take a junction at 1600 K, walls at 800 K, emissivity 0.6, and $h = 1000 \text{ W}/(\text{m}^2 \cdot \text{K})$ for a small bead in reasonable flow. The result is approximately 200 K of error. With a larger sheathed probe, where h might be 200–400 $\text{W}/(\text{m}^2 \cdot \text{K})$, the same conditions give several hundred kelvin. In mild conditions below about 700 K it shrinks to a few kelvin. This is not a defect - it is what the instrument genuinely measures.

The signatures this leaves in the data are what the software team needs to recognise:

- **Strongly nonlinear growth with temperature.** Because of the fourth-power term the error is negligible below roughly 600–700 K and grows rapidly above 1200 K. A channel that agrees with a reference at low temperature but reads progressively low at high temperature is showing radiation error, not gain drift.
- **Coupling to wall temperature.** As the rig walls heat up over minutes, radiation loss falls and the reading creeps upward even though the gas temperature is constant. A detector that flags "drift after the process stabilised" will fire on this every single run.
- **Emissivity drift from soot.** Soot deposition raises emissivity, which increases the radiation error, producing slow downward drift at constant conditions. Soot also adds insulating mass, which increases τ - so correlated drift in both level and fitted τ is a distinctive fouling signature and a genuinely useful maintenance flag.
- **Step change when flow changes.** The error scales inversely with h , so a deliberate flow change produces a step in the bias, appearing as a small unexplained level shift superimposed on the expected temperature change.

Conduction (stem) error

The probe stem is a fin connecting the hot junction to a cold mount. Modelling the immersed length L as a fin:

$$T_{tc} - T_g \approx (T_{mount} - T_g) / \cosh(mL), \quad m = \sqrt{(hP / kA_c)}$$

The error decays roughly exponentially with immersion depth. As a rule of thumb, an immersion depth of ten to twenty probe diameters makes conduction error negligible in gas; at five diameters it can be tens of kelvin in a hot-gas, cold-mount situation. The error is worst for large-diameter sheaths, low gas velocity, and cold mounts.

Its signature differs usefully from radiation error. Conduction error is a function of geometry, so it changes only when the geometry or the mount temperature changes: a step when a probe is repositioned or re-seated, typically after maintenance; slow drift as the mount heats through a run; a shift when flange coolant changes. Unlike radiation error it has no fourth-power nonlinearity, scaling linearly with the temperature difference to the mount. Instrumenting the mount flange with a cheap DS18B20 makes this directly diagnosable.

The classification lesson. Radiation and conduction errors are systematic biases of tens to hundreds of kelvin that behave exactly like slow anomalies. Two co-located probes of different diameter should disagree at high temperature - the smaller one reads closer to the true gas temperature. Cross-channel consistency checks must therefore compare like with like, or allow a temperature-dependent tolerance. But changes in these biases are worth surfacing: rising τ together with downward level drift indicates sooting; a step coincident with a maintenance window indicates a repositioned probe. Report them as diagnostics, never as grounds for rejecting data.

Velocity and recovery effects

In a fast gas stream the probe partially recovers the flow's kinetic energy as heat, reading between the static and stagnation temperatures:

$$T_{tc} \approx T_{static} + r \cdot U^2 / (2 c_p), \quad \text{recovery factor } r \approx 0.68\text{--}0.86$$

With a specific heat around 1.1–1.3 kJ/(kg·K) for hot combustion products, the effect is about 0.4 K at 30 m/s, 4 K at 100 m/s, and 35 K at 300 m/s. Below roughly 50 m/s it is sub-kelvin and can be ignored. It matters only for channels in nozzle or jet flows, where it appears as a velocity-correlated offset. The team should confirm which channels, if any, see velocities above 50 m/s.

3.4 The K-type thermocouple as a transducer

Roughly half the plausible faults live in the electrical physics rather than the thermal physics, so the detection logic needs both.

Where the EMF actually comes from

A conductor in a temperature gradient develops an EMF along its length. A thermocouple is two dissimilar wires - for K-type, chromel positive and alumel negative - joined at the measuring end. The loop voltage is the integral along the wire path of the difference in Seebeck coefficients multiplied by the local temperature gradient.

The point that is most often misunderstood. The junction generates nothing. It merely closes the circuit. The EMF is produced distributed along whatever portion of the wire lies in a temperature gradient. This has direct practical consequences for fault detection.

The most important consequence is that inhomogeneity cannot be calibrated out. If a section of wire has a locally altered Seebeck coefficient - from work hardening through bending, from oxidation, from contamination, or from K-type's two specific pathologies of short-range ordering in chromel around 250–550 °C and "green rot" chromium oxidation in low-oxygen atmospheres around 800–1050 °C - then the resulting error depends on where the temperature gradient happens to sit along the wire. Calibrating the

probe in a furnace, where the gradient is in one place, tells you nothing about its error in the rig, where the gradient is elsewhere. A probe can pass a bench check and still read wrong in service.

In the data this appears as a reading that changes when the cable is repositioned or when the gradient region shifts - insulation moved, mount heating up - with no corresponding process change. It also appears as errors confined to a specific temperature band during heat-up or cool-down. Both are legitimate grounds for flagging a probe for replacement, and neither is fixable in software.

The cold junction

The loop measures a difference between the measuring gradient region and the reference end, so the instrument must measure its own terminal temperature and add the equivalent EMF. Both MAX parts perform this compensation on-die. The failure signature is distinctive: a compensation error appears as a common-mode offset that tracks board temperature. K-type channels on the same board drifting together, correlated with a board-mounted ambient sensor, while channels on other boards stay put, indicates a cold-junction or board-heating problem rather than a process event.

This deserves specific attention on wireless nodes. The compensation sensor is on the converter die, but the actual cold junction is at the terminal block where thermocouple wire meets copper. Any thermal gradient between the two is a one-to-one additive error in the reported temperature. On a node with a radio, transmission current bursts heat the power amplifier and regulator, producing board gradients synchronised with the transmission schedule - which appears as temperature steps correlated with radio activity. Mitigations are physical: keep the converter thermally bonded to the terminal block and away from the radio and regulator, and log the cold-junction temperature in every record so this is diagnosable afterwards. The MAX31856's cold-junction fault bits should be wired directly into the detector, because compensation is a genuine single point of failure for the whole channel.

Sensitivity of about 41 $\mu\text{V}/^\circ\text{C}$: the signal is tiny

One degree Celsius corresponds to roughly 41 microvolts. Millivolt-scale interference - ground loops, igniter or motor switching, electrostatic pickup on unshielded runs, parasitic thermoelectric EMFs at connectors made of the wrong metal - therefore produces tens of degrees of artefact. The spark igniter of a combustion rig is a textbook interference source, and ignition-synchronous spikes should be treated as a recognisable, expected class of non-thermal event. This is only computable if igniter and actuator timing is logged and time-synchronised with the sensor data, which the team should arrange.

For context on resolution: the MAX31855 least significant bit is 0.25 $^\circ\text{C}$, about 10 μV . The MAX31856 least significant bit is around 0.0078 $^\circ\text{C}$, roughly 0.3 μV - below realistic wiring noise in most installations, so the extra bits are honest only with excellent layout. Expect the low bits to be noise and establish noise floors empirically rather than from the resolution figure.

Finally, every dissimilar-metal transition in the measuring circuit is a parasitic thermocouple. Extension runs must use K-type compensating cable, not copper - copper extension relocates the reference junction to wherever the transition happens, at an uncontrolled temperature.

Nonlinearity, and what it means for cross-comparison

The Seebeck coefficient for K-type is not constant. It falls off sharply below 0 $^\circ\text{C}$ and varies across the 0–1370 $^\circ\text{C}$ span, roughly within a 35–42 $\mu\text{V}/^\circ\text{C}$ band. The MAX31855 approximates this with a single straight line at 41.276 $\mu\text{V}/^\circ\text{C}$, so its error is small near the region where the line matches - roughly room temperature to a few hundred degrees - and grows toward both extremes, reaching degrees to several degrees of systematic, temperature-dependent error at combustion temperatures. The MAX31856 applies an internal lookup table and returns values already corrected to ITS-90.

Practical instruction. A systematic, reproducible divergence between a MAX31855 channel and a MAX31856 or DAQ970A channel that grows with temperature is the linearisation difference, not a fault. It is computable in advance: calculate the expected divergence curve once against the ITS-90 tables over the actual operating range, and subtract it before any cross-comparison. Take the polynomial coefficients from NIST directly rather than transcribing them from secondary sources.

Junction style

The choice of junction changes both priors the detector relies on - the time constant and the expected noise signature - so it must be a per-channel configuration field, and it should be verified rather than assumed. An ohmmeter check between lead and sheath at commissioning distinguishes grounded from ungrounded; mixed stock is common.

Junction type	Relative τ	Electrical behaviour	Detection-relevant consequence
Exposed (bead in flow)	Fastest	No isolation; the bead is effectively an antenna and corrodes or soots directly	Highest interference susceptibility and fastest fouling drift. The only option for fast flame dynamics.
Grounded (welded to sheath tip)	2–3× faster than ungrounded of the same diameter	Junction galvanically connected to sheath and hence to rig metalwork	Ground loops; common-mode currents through the rig appear in signal; multiple grounded probes can couple to each other.
Ungrounded (isolated in MgO)	Slowest	Electrically isolated until the insulation degrades	Cleanest electrically. Its high-temperature insulation resistance is precisely what fails in the virtual-junction mode described below.

The most dangerous single failure mode: shunt impedance / virtual junction. Above roughly 800–1000 °C, ceramic and fibreglass insulation loses resistivity by orders of magnitude. The two legs effectively short along the wire, inside the hot zone, and the EMF is then generated at that new junction rather than at the tip. The reading is stable, noise-free, physically plausible - and wrong. Open-circuit detection cannot catch it, because the circuit is not open. The only reliable indicator is that the apparent time constant changes, because the effective junction now sits in a different thermal environment. This is the strongest single argument for implementing in-situ τ tracking.

4. Turning the physics into detection tests

This is the core translation. Every test below is computable from the first minute of the first run of a brand-new experiment, with no historical data of any kind.

4.1 Amplitude bounds from thermodynamics

The environment cannot present temperatures outside the range of available sources and sinks.

- **Upper bound.** The adiabatic flame temperature of the fuel and oxidiser in use - around 2200–2400 K for stoichiometric hydrocarbon-air, higher for oxygen-enriched or hydrogen systems. The operating team should supply this per rig configuration. A reading above the maximum achievable source temperature is a fault without qualification. The sensor's own limits apply too: K-type is usable to about 1370 °C, and the converter's linearisation range and the wire's melting point bound what can possibly be real.
- **Lower bound.** The coldest available sink - ambient, or coolant inlet temperature. Nothing passive reads below it.
- **Per sensor class.** A DS18B20 tops out at +125 °C and its ± 0.5 °C specification holds only between –10 and +85 °C, degrading toward ± 2 °C at the extremes. SHT45 and DHT11 are ambient-class devices. Plausible windows are per channel class, never global.

4.2 Rate bounds and the sensor as a low-pass filter

The slope bound from Section 3.2 gives an immediate per-channel test. Numerically: a bare 0.25 mm bead with τ of 10 ms and a maximum drive of 2400 K permits enormous slopes - effectively unbounded at practical sample rates, so for that channel the frequency-domain test below does the real work. A sheathed 3 mm probe with τ of 15 s permits at most about 140 K/s, and far less once the reading is near process temperature; such a channel jumping 50 K between samples 100 ms apart is provably reporting something non-thermal.

The deeper statement is that a first-order lag is a single-pole low-pass filter:

$$|G(f)| = 1 / \sqrt{1 + (f/f_c)^2}, \quad f_c = 1/(2\pi\tau)$$

Corner frequencies follow directly: $\tau = 20$ ms gives about 8 Hz; $\tau = 1$ s gives 0.16 Hz; $\tau = 20$ s gives 0.008 Hz. Above the corner the physical world is attenuated at 20 dB per decade, so two decades above it genuine thermal content is down by 40 dB.

The spectral test. Signal energy at frequencies well above the channel's corner, in excess of the known electrical noise floor, is definitionally not temperature. Sustained energy at mains frequency or its harmonics is pickup; broadband high-frequency energy is interference or a digital-chain problem; a single-sample excursion is a communications or converter event.

Implementation is cheap: a rolling power spectrum, or even a simple high-pass-filtered RMS per channel with the cutoff set a decade above that channel's corner, gives a training-free non-thermal energy monitor. Establish each channel's electrical noise floor by recording a minute of data with the probe at stable ambient before the run - that is a reference level, not a training corpus.

Two cautions. First, turbulent combustion genuinely fluctuates at tens to hundreds of hertz, and a fine-wire bead legitimately passes some of that while a sheathed probe cannot - so the "impossible band" must be set per channel from that channel's τ . Second, the MAX31856's built-in mains filter will suppress pickup on its channels while the MAX31855 has no such filter, so identical probes on the two front-ends will legitimately show different noise signatures.

4.3 The two-sensor lag check

This is the single most powerful zero-baseline cross-check available, and it is inexpensive. If two sensors with different time constants watch the same process, both are low-pass filters of the same unknown input. Without ever knowing the true gas temperature, the slower sensor's reading must be approximately the faster sensor's reading passed through a first-order filter of the slower time constant:

$$\hat{y}_{\text{slow}}[k] = \alpha \cdot \hat{y}_{\text{slow}}[k-1] + (1 - \alpha) \cdot y_{\text{fast}}[k], \quad \alpha = \exp(-\Delta t / \tau_{\text{slow}})$$

Run that predictor and monitor the residual. Its expected behaviour is fully specified by physics:

- **The fast sensor must always lead.** Every genuine transient appears on the small- τ channel first and on the slow channel as a delayed, amplitude-reduced version. If the slow channel moves first, or moves alone, the event on it is not thermal - or the two sensors are not actually watching the same process, which is also worth knowing.
- **The residual should be small and zero-mean during transients,** apart from the systematic bias differences of Section 3.3, which are slow and level-dependent and can be removed with a slowly adapting offset.
- **A growing lag mismatch means a time constant has changed** on one of the two probes - fouling or degradation. Fitting each independently identifies which.

The same logic extends loosely across sensor types: a DS18B20 on the rig casing must respond to a burner event after, and more slowly than, an in-stream thermocouple, with a lag consistent with the conduction path between them. Even coarse ordering constraints of the form "A before B before C" are physics-derived and need no training data. Where the rig design allows it, deliberately pairing a fast and a slow probe at key locations is worth recommending as an instrumentation practice, precisely because it buys this check.

4.4 Measuring τ in situ, and reading its drift

τ should be measured rather than assumed. At commissioning, apply a step - plunge the probe into a stirred bath, or simply use burner ignition or shutdown - and fit the logarithm of the temperature difference against time; the slope gives τ . Fit the portion after roughly the first 10 % of the transient to avoid the two-lump early dynamics of sheathed probes. Store τ per channel together with the fit uncertainty. Thereafter, every natural transient in normal operation is a free experiment: re-fit opportunistically and trend the result.

A drift in fitted τ is itself a first-class diagnostic, and it is one of the few ways to detect failures that leave the reading looking entirely plausible.

Change in fitted τ	Likely physical cause
Grows gradually, with level drifting down at high temperature	Soot or fouling layer on bead or sheath - added mass, added insulation, and raised emissivity acting together
Grows after cold or humid operation	Condensate on the sheath, or in the probe or connector
Drops suddenly, possibly with a level shift	Shunt impedance / virtual junction failure - insulation resistance has collapsed at high temperature and the effective junction has moved along the cable. Dangerous because the reading stays plausible.
Apparent $\tau \approx 0$, instantaneous steps	Not a thermal signal at all - electrical or digital in origin

4.5 Trajectory and energy-balance sanity

Beyond instantaneous checks, whole trajectory shapes are constrained by physics.

- **No source, no rise.** With burners off and no exothermic process running, no channel may rise above its neighbours or ambient beyond noise. A rising reading in a dead rig is either a fault or an unnoticed energy source - surface it either way.
- **Free cooling is monotone and exponential toward the sink.** After shutdown every channel must decay monotonically toward ambient or coolant temperature, well described by one exponential for a bare bead or two for a sheathed probe. Note that what the sensor tracks during cooling is the rig's own, slower thermal relaxation, so the observed decay constant is the rig's - but it cannot be faster than the sensor's τ permits. Non-monotonic excursions beyond noise, plateaus that break the fit, or decay faster than the sensor allows are faults. The shutdown tail of every run is therefore a built-in self-test that costs nothing to exploit.
- **Convergence under cooling.** Everything converges toward the same ambient. Channels crossing each other repeatedly during quiet cooling indicates a channel problem.

4.6 A statistical layer, correctly bounded

Statistical methods have a place, but a strictly subordinate one: they annotate, they do not veto, and their parameters should be anchored to datasheet or physical values rather than tuned by hand.

Residual monitoring

A predictor whose memory is set by the sensor's own time constant - rather than by an arbitrary tuning constant - produces a residual that can be monitored against an adaptive scale estimate. Three design points matter more than the specific estimator chosen:

- Anchor the filter's memory to τ . This removes the free parameter and makes the filter able to follow every transient the sensor is physically capable of producing, and nothing faster.
- Limit the influence of any single sample on the filter's rate estimate. Without this, one large communications glitch injects a spurious rate that causes a cascade of false alarms for several time constants afterwards - a common and confusing failure of naive implementations.
- Freeze the scale estimate while an alarm is active. Otherwise a sustained fault progressively inflates the estimated noise scale until the fault is absorbed as normal and the detector goes blind to it.

The lower bound on the noise scale should come from the datasheet rather than from data. Quantisation noise alone is the resolution divided by $\sqrt{12}$ - about 0.072 °C for a MAX31855 at 0.25 °C resolution, about 0.018 °C for a DS18B20 at 12-bit, 0.036 °C at 11-bit, 0.144 °C at 9-bit. The converter's own input-referred noise adds to this in quadrature and should be checked against the datasheet, though in practice on a real rig the total is usually dominated by interference rather than by the converter.

Drift detection

Running a drift detector continuously on a live experiment will alarm constantly, because the process is supposed to be changing - that is what an experiment is. Drift detection must therefore be gated: compute one statistic per validated quiescent window, where a window qualifies as quiescent only if its range is small relative to the noise floor, the fitted rate is near zero, and no quality flags were raised within it. A cumulative-sum or exponentially-weighted control chart then runs on that decimated series, at perhaps one value every five minutes rather than one per second.

Thresholds can be set analytically rather than by trial and error. Standard control-chart theory relates the reference value and decision interval to the average run length between false alarms, and the shift worth detecting can be derived from physical tolerance - for instance, some fraction of the IEC 60584 tolerance

class of the thermocouple in use. Verify the current edition of that standard for the exact tolerance formulas.

The reference against which drift is measured should be chosen in this order of preference: a cold-soak self-check where the thermocouple and cold-junction readings must agree when the rig is off and equilibrated, which is the only check available that touches absolute accuracy rather than self-consistency; differences against sibling channels during quiescent windows; the same operating point repeated across runs; and the trend in fitted τ .

What not to build

Unsupervised machine learning on the node - autoencoders, one-class classifiers, isolation forests, sequence predictors - should be avoided, for the reasons in Section 1.1: without labels there is no way to validate them, without a stationary normal they flag every novel condition, and they absorb pre-existing faults as normal. The legitimate role for adaptive computation is gray-box parameter estimation: fitting a handful of physically interpretable parameters per channel, each of which can be inspected, plotted across runs, and reasoned about. Time constant, noise scale, quiescent offsets between channels, and the cold-soak offset are all parameters of a physical model rather than learned decision boundaries. That distinction is what makes them acceptable in a research instrument.

A sensible division of labour. Since a cloud analytics layer already exists, let it do what it is good at: fitting parameters across many runs and many nodes, where there is enough data and compute to be trustworthy. Push those parameters down to the node as a versioned configuration. The node then executes deterministic, auditable rules in real time, works offline, and reacts in milliseconds rather than seconds. Stamp the parameter-set version into every record so any dataset can be reproduced exactly.

5. Validating against the DAQ970A dataset

Using recorded Keysight DAQ970A data as the test set for the detection software is sound, but the reference instrument is not a transparent window onto truth. Three properties determine how validation results must be interpreted.

5.1 NPLC integration makes the instrument a different low-pass filter

The DAQ970A integrates its input over a whole number of power-line cycles. On 50 Hz mains, 1 NPLC is 20 ms of integration per reading, 10 NPLC is 200 ms, and 100 NPLC is 2 s. Integration over a window is a moving-average filter whose frequency response has nulls at multiples of the reciprocal of the integration time - which is precisely why integrating over whole line cycles annihilates mains pickup. That is the feature. The side effect is that the effective bandwidth is roughly 0.44 divided by the integration time: about 22 Hz at 1 NPLC, 2.2 Hz at 10 NPLC, and 0.22 Hz at 100 NPLC. At the high-NPLC settings that deliver the headline resolution, the instrument is a sub-hertz low-pass filter before the sensor's own time constant is even considered.

Auto-zero, if enabled, adds a zero measurement per reading, roughly doubling the per-reading time. With the 34901A multiplexer scanning several channels, each channel is sampled at the aggregate rate divided by the channel count, and channels within a scan are sequential rather than simultaneous - introducing inter-channel skew that can reach hundreds of milliseconds.

Consequences for validation. Fast artefacts are invisible in the reference data. A 5 ms interference spike or a single corrupted SPI word simply does not exist in a 200 ms-integrated record. The DAQ970A dataset can validate the slow physics checks - bounds, drift, cooling curves, cross-channel lag at slow timescales - but it cannot certify spike or glitch detection. Those must be tested against node hardware with deliberately injected faults.

It follows that thresholds do not transfer between the two. Sample-to-sample noise, maximum observed change per sample, and spectral content all differ between a DAQ970A record and a free-running MAX31855 or a MAX31856 with its own filtering and averaging configuration. An algorithm whose constants were tuned on the smooth reference data will false-alarm on node data; tuned the other way, it will be blind. Every threshold must be parameterised by the channel's time constant, its sample interval, and its front-end noise floor - never baked in as a constant derived from one instrument.

One further subtlety: at 100 NPLC the instrument's own bandwidth of 0.22 Hz can be below the corner frequency of a fast probe. The reference then under-reports genuine fast thermal content, and a node channel showing more high-frequency energy than the DAQ970A record is correct, not anomalous.

5.2 Accuracy versus repeatability

Two different quantities are both called accuracy, and they play opposite roles here.

- **Absolute accuracy** - how close the reported value is to true temperature. For a thermocouple channel this budget is dominated by the probe, not the instrument. IEC 60584 Class 1 wire is specified at $\pm 1.5^\circ\text{C}$ or $\pm 0.4\%$ of reading, whichever is larger, which is $\pm 6^\circ\text{C}$ at 1500°C ; Class 2 is $\pm 2.5^\circ\text{C}$ or $\pm 0.75\%$. And that is before the installation biases of Section 3.3, which can be one to two orders of magnitude larger. The instrument's own contribution to a thermocouple measurement is of order 1°C - the frequently quoted 0.003 % figure refers to DC voltage mode, not to the end-to-end temperature chain. Check the exact thermocouple-mode specification at your settings.
- **Repeatability, or noise floor** - how much the reported value wobbles when the true value is constant. Here the instrument at high NPLC is excellent, at the level of hundredths of a degree.

The distinction that matters. Anomaly detection consumes repeatability, not absolute accuracy. Every check in Section 4 examines changes, rates, residuals and spectra - differential quantities in which a constant offset cancels. Therefore a steady level offset of a few degrees between the DAQ970A record and a node channel reading the same probe is expected, not a detection failure. Compare dynamics, not absolute levels; or remove a fitted constant offset before comparing. Equally, do not present the reference as giving truth to hundredths of a degree - it gives stability at that level, but its truth is only as good as the probe class and installation biases, exactly like everyone else's.

5.3 Practical checklist for using the validation set

1. Record and store alongside the data: NPLC setting, auto-zero state, scan list, per-channel timestamps rather than only scan-start time, probe type, tolerance class, junction style, and insertion depth. Without this metadata the dataset becomes uninterpretable within months.
2. Time-align node and instrument records explicitly. Multiplexer skew across a scan can exceed the duration of the transients of interest.
3. Validate the slow physics checks against the DAQ970A data - bounds, drift, cooling-curve fits, cross-channel lag, τ estimation. These are exactly what the reference is good for.
4. Validate the fast checks against node hardware with injected faults: pull a connector mid-run, fire the igniter, deliberately corrupt a bit in software, disconnect a 1-Wire device. The reference instrument cannot see any of these, so they need a separate test campaign.
5. Precompute and subtract the expected MAX31855 linearisation divergence before any level comparison, so a known systematic difference is not scored as a detection.

6. Summary: physics to detector

Each row is a test that requires no historical baseline and no labelled data.

Physical principle	Test to implement	What a violation means
First-order lag, $\tau = \rho V c / (hA)$	Per-channel slope limit: $\Delta T / \Delta t \leq (T_{\text{drive,max}} - T) / \tau$, with margin for τ uncertainty	Non-thermal: communications, interference, converter, or software fault. Recall the MAX31855 has no CRC.
Sensor is a low-pass filter, $f_c = 1/(2\pi\tau)$	High-pass RMS or spectral energy above roughly ten times f_c , referenced to a pre-run noise floor	Electrical pickup: mains frequency and harmonics, broadband interference, or single-sample digital events
Thermodynamic bounds	Reading within [coldest sink, hottest source] for that channel class	Fault, unconditionally
No energy source means no temperature rise	Monotonicity check in a dead rig	Fault, or an unnoticed energy source - surface either way
Free cooling is monotone exponential	Fit the shutdown tail to an exponential decay	Fault. Every run ends with a free self-test.
Two sensors, different τ , same process	Predict the slow channel from the fast one; monitor residual and lead/lag ordering	Wrong ordering means a non-thermal event; a drifting fit means one τ has changed
τ is measurable in situ and should be stable	Re-fit τ opportunistically on every transient and trend it	Rising τ indicates fouling or condensate; falling τ with a level shift indicates virtual-junction failure; τ near zero means electrical
Radiation error $\propto \epsilon \sigma (T^4 - T_w^4) / h$	Expect nonlinear high-temperature bias, wall-tracking creep, soot-correlated drift	Not a fault. Annotate as a diagnostic; never reject the data.
Conduction error is a function of geometry	Expect steps on repositioning and drift with mount temperature; instrument the mount	Not a fault. A step with no maintenance event warrants investigation.
EMF arises in the gradient region, at $\sim 41 \mu\text{V}/^\circ\text{C}$	Correlate artefacts with igniter and actuator events; watch for band-specific errors on heat-up	Inhomogeneity or interference. Inhomogeneity is uncalibratable - flag the probe for replacement.
Cold-junction compensation is a shared single point of failure	Watch for channels on one board drifting together with board temperature	Compensation or board heating, not a process event
Physics may veto; statistics may only annotate	Classify every flag as implausible, suspicious, or unusual	Protects genuine discoveries from being silently suppressed

Suggested implementation order

If the work is to be phased, this ordering front-loads the checks that catch silent, unrecoverable failures at the lowest implementation cost.

1. **Signal integrity.** Structural validation of the MAX31855 word, MAX31856 fault register decoding, DS18B20 CRC validation, and ROM-ID-based addressing. These catch failures that produce plausible-looking wrong numbers, which nothing downstream can recover from.
2. **Datasheet and thermodynamic bounds.** Per-channel range limits and sensor-class plausibility windows. Trivial to implement, and immediately useful.
3. **Per-channel τ configuration and the slope bound.** Requires commissioning measurements, but delivers the strongest single physics-based test.

4. **Cold-soak self-check and the quality-flag vocabulary.** Near-zero cost, and the only check that touches absolute accuracy.
5. **Spectral and within-interval scatter monitoring.** Catches interference and mechanical connection problems early.
6. **Two-sensor lag consistency, where probe pairs exist.** High diagnostic value for modest effort.
7. **In-situ τ fitting and trending.** The only reliable route to detecting virtual-junction failure and fouling.
8. **Gated drift detection.** Valuable but must be built on the quiescent-window gating, or it will produce constant false alarms.

Items to verify rather than assume

- Per-channel τ - measure at commissioning by plunge test or ignition step, then re-fit online. Do not use catalogue values.
- Maximum and minimum drive temperatures per rig configuration, from the combustion team.
- Junction style for each probe, confirmed by ohmmeter check between lead and sheath; also insertion depth and tolerance class.
- Which channels, if any, see gas velocities above 50 m/s, for recovery-effect correction.
- MAX31855 reserved-bit positions and conversion timing; MAX31856 filter and averaging configuration, fault-register bit map, and common-mode specifications - all from the current datasheets.
- DS18B20 self-heating figure for the specific package, mounting and duty cycle in use.
- DAQ970A capture settings for every validation dataset: NPLC, auto-zero state, scan interval, per-channel timestamps, and the instrument's thermocouple-mode accuracy specification at those settings.
- MAX31855 linearisation error curve across the actual operating range, computed once against the NIST ITS-90 tables.
- IEC 60584-1 tolerance formulas for the thermocouple classes in use, from the current edition of the standard.
- Rig event log - igniter, valves, pumps - time-synchronised with sensor data. Many apparent anomalies are explainable actuator interference, and that correlation is only computable if the events are recorded.

Closing note for the implementation team. The recurring theme across every section is that the most dangerous failures in this hardware are the ones that produce plausible numbers: an all-zero SPI word reading as 0.00 °C, a 1-Wire device dropping out and silently relabelling every downstream channel, an insulation breakdown relocating the measuring junction inside the cable. None of these look like outliers. They are caught structurally - by integrity checks, by identity verification, and by asking whether the signal obeys the physics of a heated object - not by statistical distance from a mean.